# Dmitry Matveev

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## Education

### Northeastern University, Boston, MA

Master of Science in Computer Science — May 2025

**Relevant Courses**: Database Management, Mobile App Development, Natural Language Processing, Foundations of Artificial Intelligence, Web Development. — GPA: 3.8

Bachelor of Science in Computer Engineering and Computer Science — May 2024

**Honors**: Dean's List, Magna Cum Laude — GPA: 3.7

**Relevant Courses**: Computer Architecture, Object Oriented Design, Algorithms, Software Engineering.

**Clubs**: AeroNU

## Work Experience

**Wellington Management** — Feb. 2022 – Jul. 2022

Core Transactional Technology Co-op

- Built a test result reporting system, later integrated into the company database.
- Developed various small-scale scripts for internal team use.
- Collaborated with another team to add web features to internal company applications.
- Implemented new data processing and visualization methods.

## Projects

**Natural Language Inference Model Comparison** — Jan. 2025 - May 2025

- Implemented Parameter-Efficient Fine-Tuning techniques including prefix tuning and Low-Rank Adaptation to optimize LLMs for 3-way classification tasks with minimal trainable parameters.
- Developed model quantization solutions for Gemma 2B using 4-bit weights with higher precision computational formats (bfloat16) to overcome CUDA memory constraints on consumer hardware.
- Designed and evaluated advanced prompt engineering strategies including Chain-of-Thought and Tree-of-Thought prompting with GPT-4o to improve zero-shot inference capabilities.

**Underwater Acoustic Transducer Geolocation System** — Jun. 2023 – May 2024

- Designed and implemented a robust navigation system for unmanned underwater vehicles (UUVs), using synchronized acoustic beacons and multilateration techniques inspired by GPS architecture.
- Developed localization algorithms combining Time-of-Flight measurements with Bayesian inference to track UUV positions in underwater environments.
- Created signal processing pipelines for underwater acoustic communications, implementing OFDM and BPSK modulation schemes with DSP filtering for noise reduction and channel estimation.

**Sacred Syntax: Global Language Detection Model** — Jan. 2024 - May 2024

- Designed and implemented dual-approach language classification system combining Support Vector Machines and Neural Networks to identify languages from textual inputs.
- Developed comprehensive text vectorization pipelines using both character-based n-grams and bag-of-words techniques to capture distinctive linguistic features.
- Engineered a recurrent neural network architecture with custom preprocessing layers to effectively process variable-length input sequences for language classification.

**Covey Town: Authentication Extension** — Sep. 2023 - Dec. 2023

- Designed and implemented a relational database system using PostgreSQL and Prisma ORM to enable user data persistence in a virtual environment application, following RESTful API principles.
- Developed secure user authentication with Auth0 integration, implementing proper callback mechanisms while adhering to security best practices for credential management.
- Engineered backend architecture using TypeScript, implementing controller-service pattern with TSOA for API route generation and endpoint isolation to ensure maintainable and extensible code.

**Technical Skills**: C/C#, Java, Python, SQL, R, TypeScript.